

Instruction Scheduling:

- ⇒ Instruction scheduling is a compile time activity that tries to improve the quality of the code that the compiler produces.
- ⇒ Like most optimizations, the scheduler runs at compile time. It analyzes the code and reorders the operations before they are emitted by the compiler.

Global Code Scheduling:

- o In the fifth phase of compiler design, code optimization is performed.
- o There are various code optimization techniques.
- o But the order of execution of code in a computer program also matters in

Date: _____

Day: M T W T F S S

Code optimization.

⇒ Global code scheduling in compiler design is the process that is performed to rearrange the order of execution of code which improves performance.

⇒ it comprises the analysis of different code segments and finding out the dependency among them.

Goals of Code Scheduling:

- Optimize the execution order
- Improving the performance
- Reducing the idle time
- Maximize the utilization of resources.

Techniques to perform Global Code Scheduling:

- Primitive code motion
- Upward code motion ○ Downward code motion

Date: _____

Day: M T W T F S S

Global Scheduling Algorithm

- Trace Scheduling
- List Scheduling
- module scheduling
- Software Pipelining

Q. What is list scheduling?

- List scheduling is a technique used for scheduling instructions in a way that optimizes performance.
- There are various types of list scheduling algorithms, including:
 - Basic Block Scheduling
 - Trace Scheduling
 - Super block scheduling
 - Loop Scheduling
 - Software Pipelining
 - Global Scheduling
 - Forward list scheduling
 - Backward list scheduling

Q. What is local list scheduling?

⇒ Local list scheduling is a type of instruction scheduling technique in compiler construction that focuses on scheduling instructions within a limited scope, typically a basic block or a small code region.

⇒ Local list scheduling confines its optimization efforts to a smaller unit of code.

Key characteristics of local list scheduling:

- Basic block scope
- Data dependency consideration
- Limited code region
- Sequential Order

Q. Explain functions that two pass assembles perform?

o The two pass assembles performs the following functions.

o it performs some function in pass 1 and some function in pass 2.

Pass 1

o Assign addresses to all statements in the assembly language program.

o Save the addresses with label for use in pass 2.

o Define symbols and literals

o Determine the length of the machine instructions

o Keep track of location counters.

o Provides some assembly instructions or operations.

Part 2

- o Create the object program
- o Perform processing of assembler directives which are not done during the pass 1.

